

Ensemble Methods

Boosting vs Bagging

From-scratch Decision Tree, AdaBoost, Random Forest, PCA, K-Means, and DBSCAN

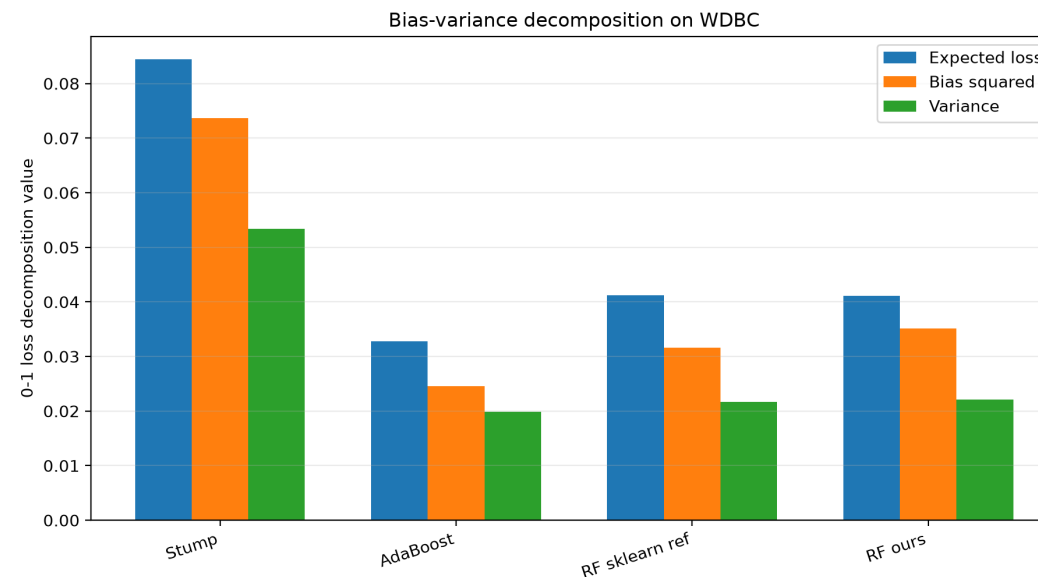
AI Academy · National AI Center · Spring 2026

When does boosting outperform bagging, and when is bagging safer?

Boosting learns sequentially from errors; it can reduce bias but may chase mislabeled samples.

Bagging trains independent models on bootstrap samples; it mainly reduces variance.

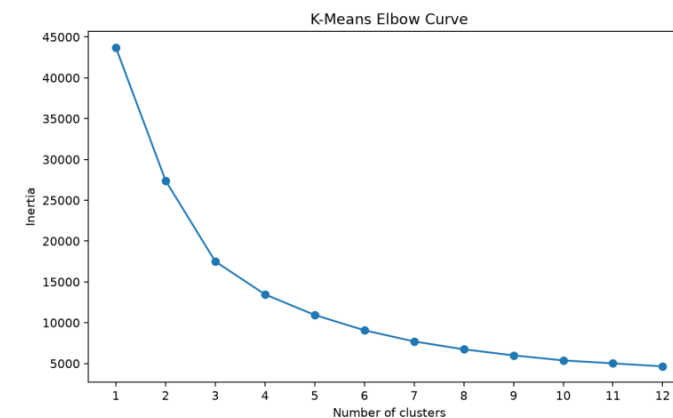
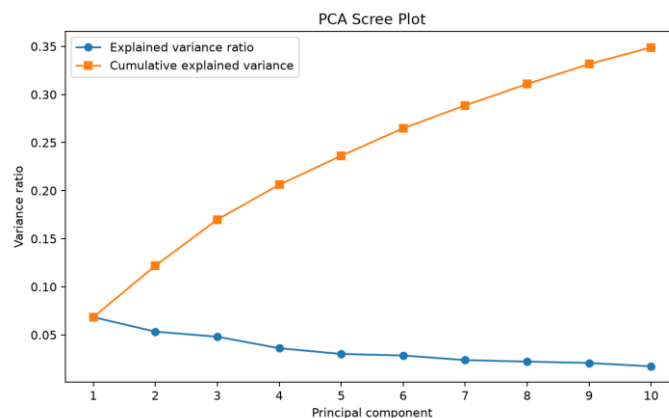
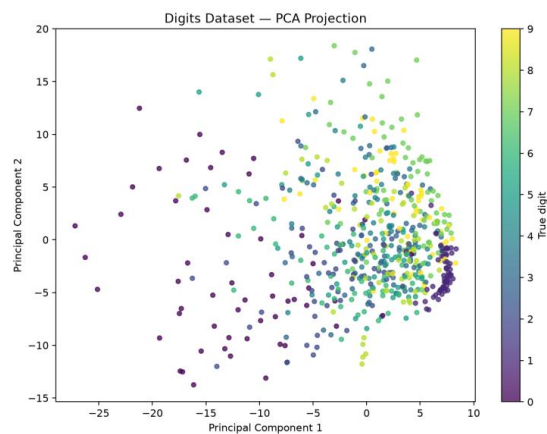
The comparison must consider noise, imbalance, dimensionality, runtime, and stability.



Core defense line: boosting lowers bias; bagging lowers variance.

Local data files prepared by download scripts: WDBC, Adult, Covertypes, MNIST 3-vs-8.
Train/test splits use fixed random_state and stratification.
Scaler is fitted on training data only to avoid leakage.
Raw dataset files are reproducible and should not be committed.

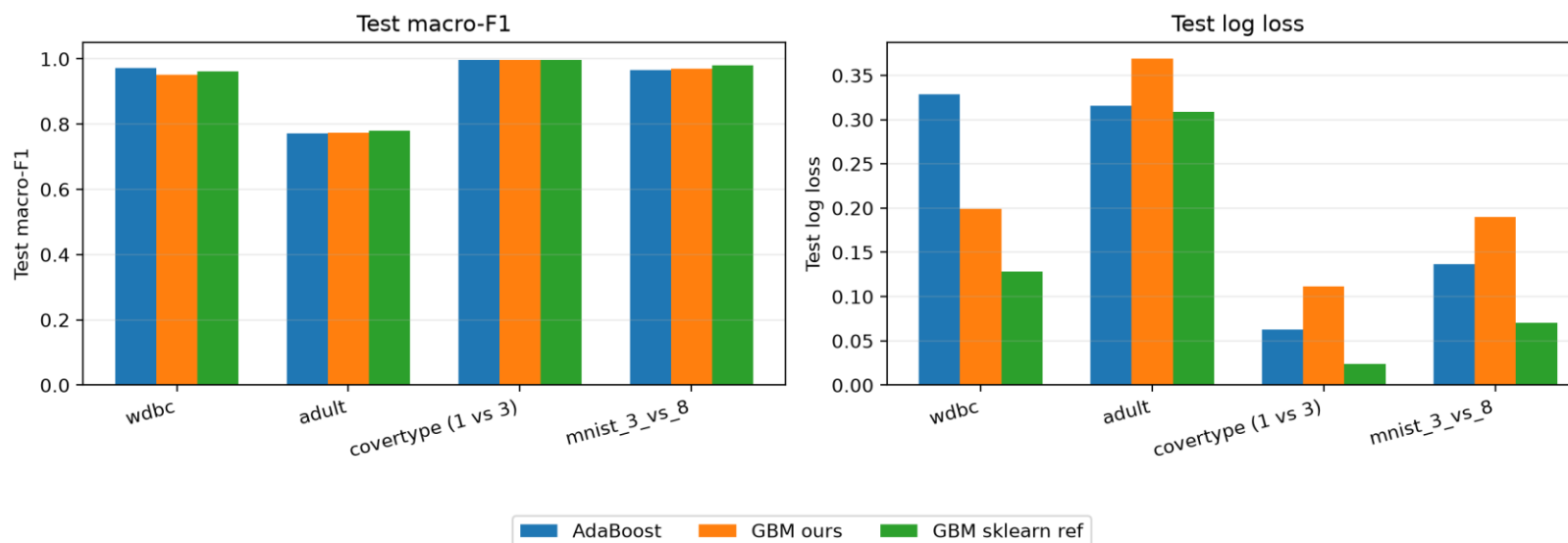
Dataset	Task	Size / features
WDBC	binary	569 / 30
Adult	binary	5,000 / 14
Covertypes	multi/imb.	5,000-10,000 / 54
MNIST 3-vs-8	binary	5,000 / 784



trees/: CART Decision Tree, impurity, splitter, stopping logic.
boosting/: AdaBoost, SAMME/SAMME.R, Gradient Boosting bonus.
bagging/: RandomForestClassifier with OOB and multiprocessing.
unsupervised/: PCA, K-Means, DBSCAN.
experiments/: run_all, scaling, noise, CV, bias-variance, QA.
tests/: 151 passed, 1 skipped in latest local verification.

All main ML algorithms are implemented from scratch.
scikit-learn is used only for utilities, metrics, splitting, and references.
Deliverables include code, tests, report, slides, notebooks, figures, and contribution report.

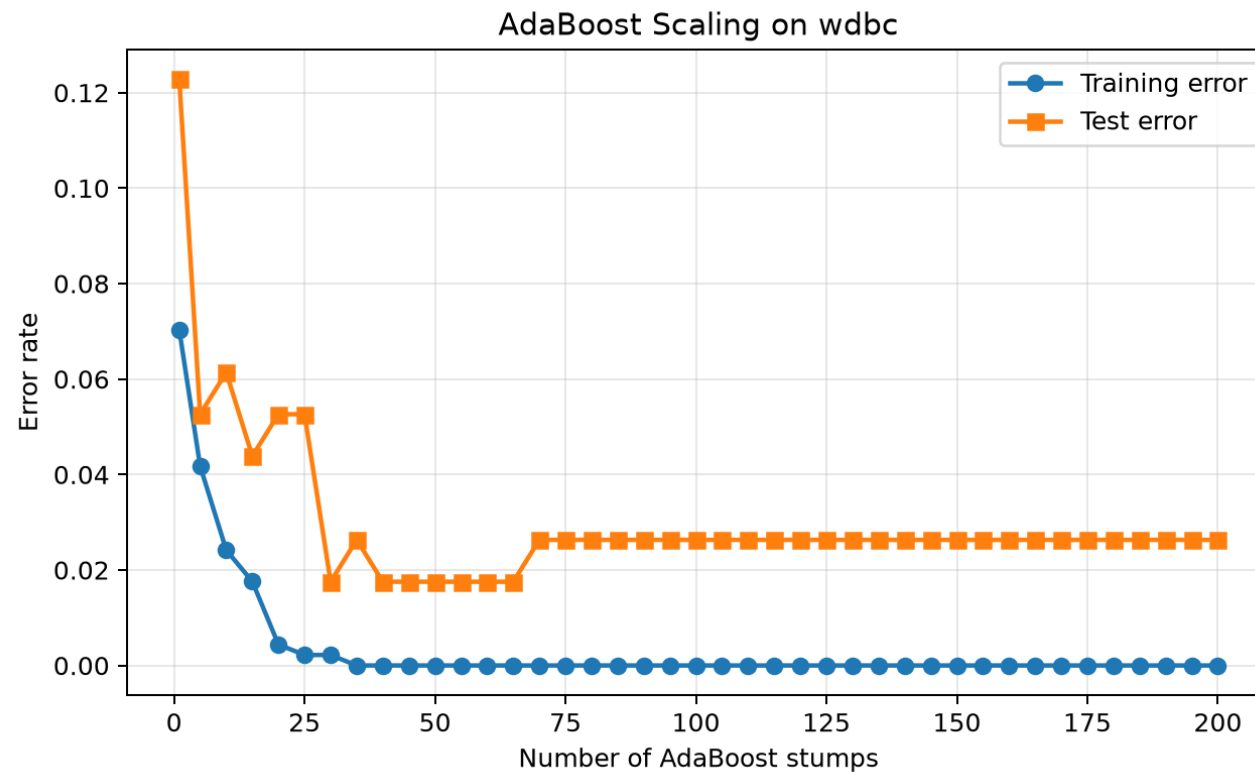
Gradient Boosting comparison on project datasets



Decision Tree baseline validates the foundation

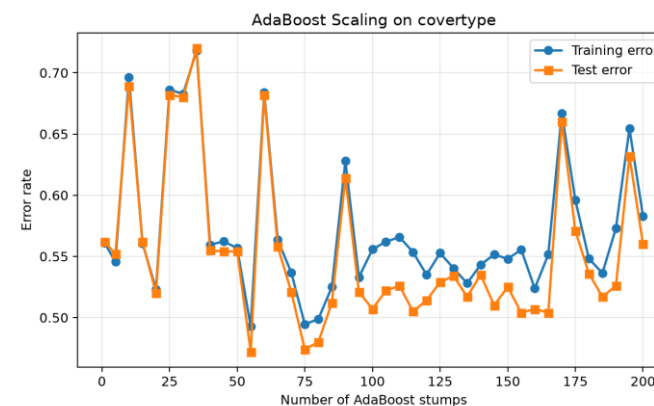
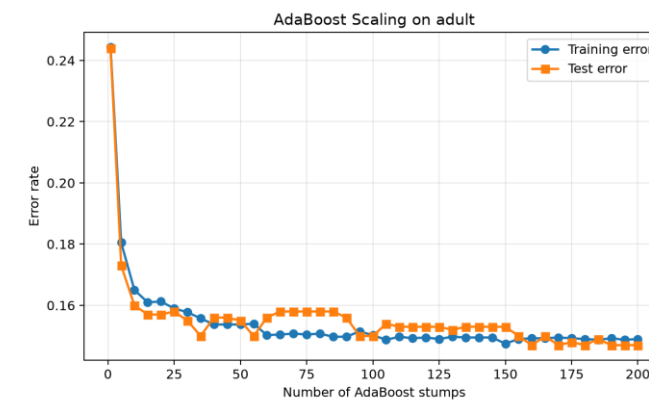
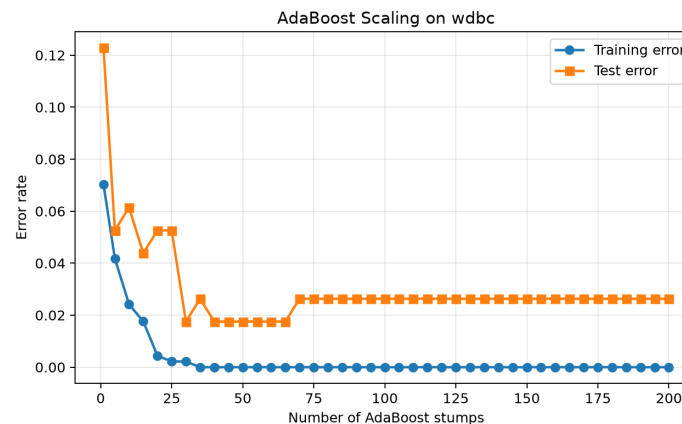
CART-style binary split: $x_j \leq \text{threshold}$.
Criteria: Gini and entropy.
Best split maximizes weighted impurity reduction.
Sample-weight support enables AdaBoost.

Model	Acc	Macro F1	AUC
MyTree	0.930	0.924	0.920
MyStump	0.877	0.860	0.843
sklearn ref	0.930	0.925	0.925



AdaBoost scaling: sequential error correction

Decision stumps are weak learners.
Misclassified samples receive larger weights.
Estimator vote weight depends on weighted error.
Strength: clean structured data. Risk: noisy labels.



Person 3 focus: Random Forest internals

~63.2%

unique in-bag rows

~36.8%

out-of-bag rows

$\sqrt{p}$

features per node

hard vote

predict()

Each tree is trained on a bootstrap sample with replacement.

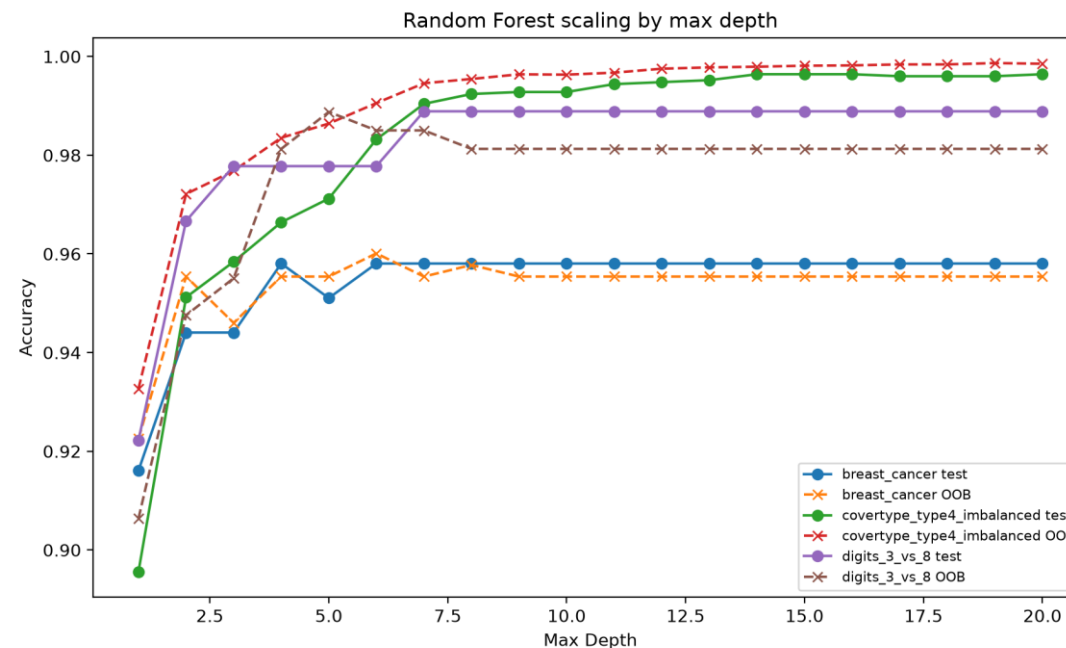
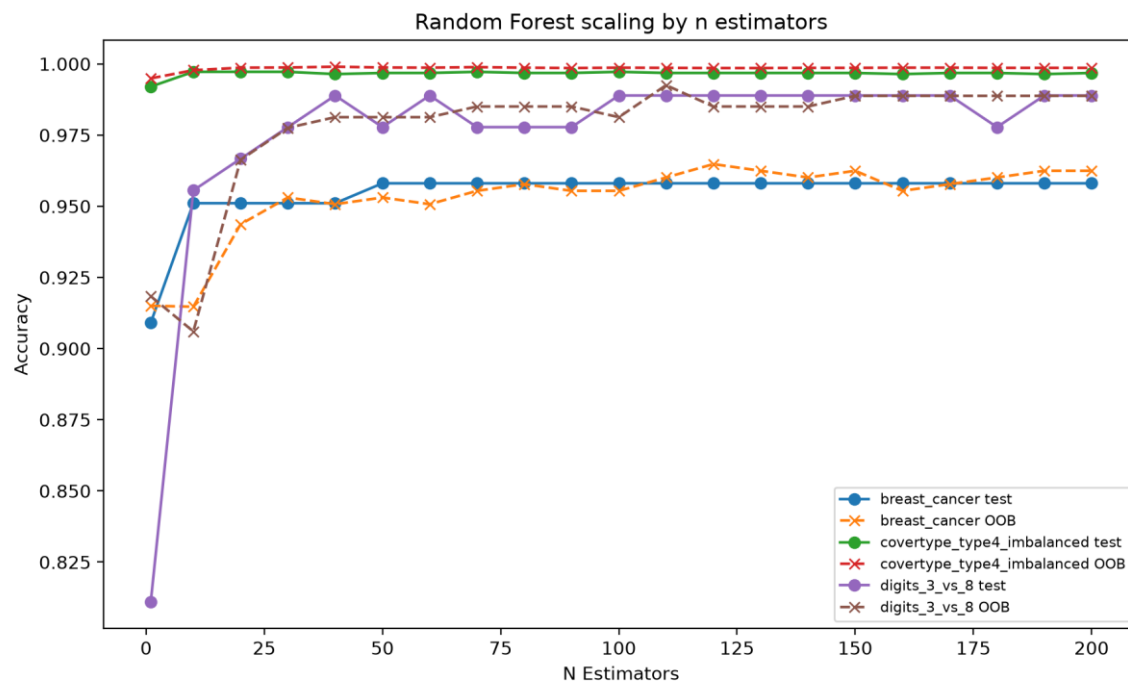
Every node searches only a random subset of features to reduce tree correlation.

OOB score uses trees that did not see a sample during training.

Key QA edge case: a tree may miss a rare class; probabilities are aligned to global classes.

Implementation: private CART tree + RandomForestClassifier + process-based multiprocessing.Pool.

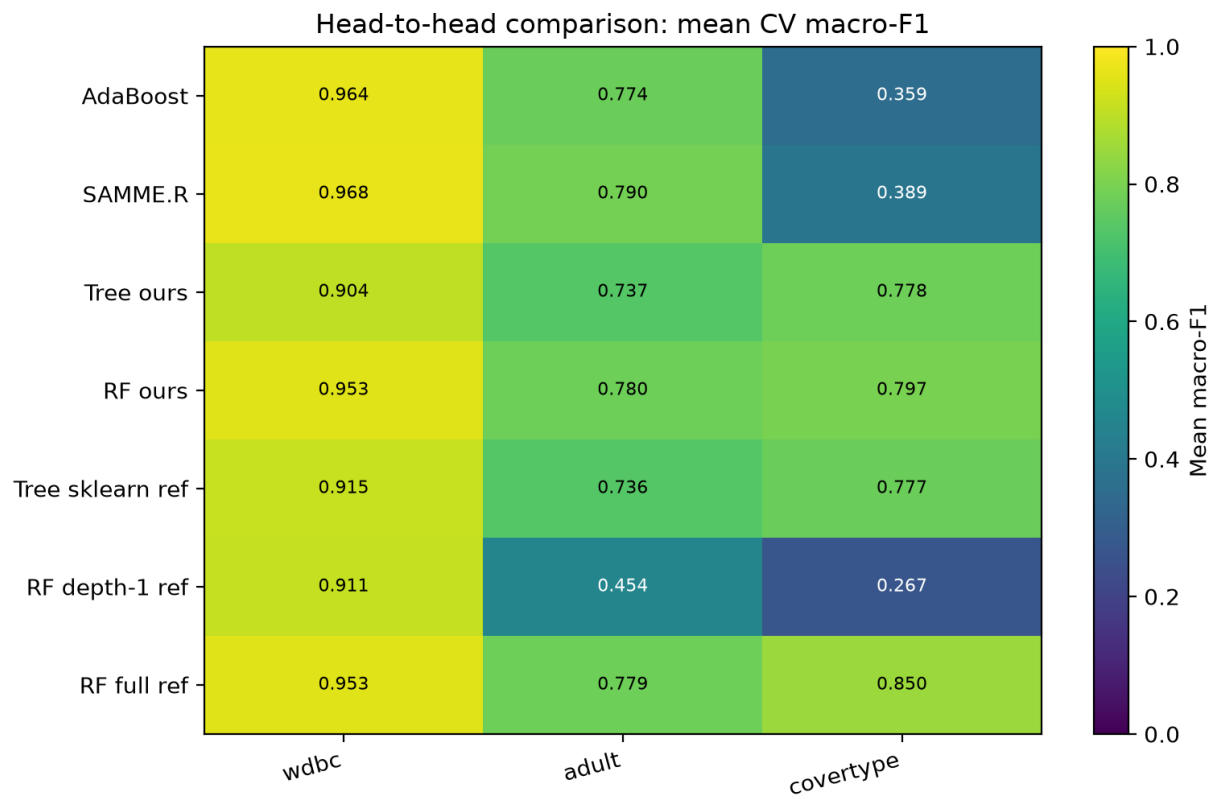
Random Forest scaling: estimators and depth



Increasing trees stabilizes OOB/test curves; depth controls underfitting vs. individual-tree variance.

Head-to-head CV: fixed estimator resources

Comparison



Dataset	AdaBoost	RF
WDBC	0.967	0.956
Adult	0.848	0.849
Coverytype	0.483	0.829

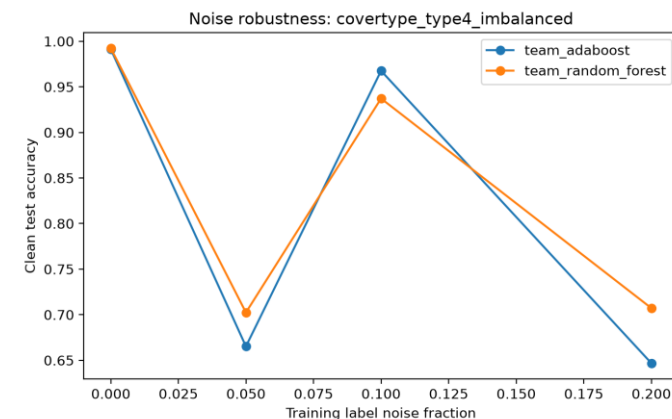
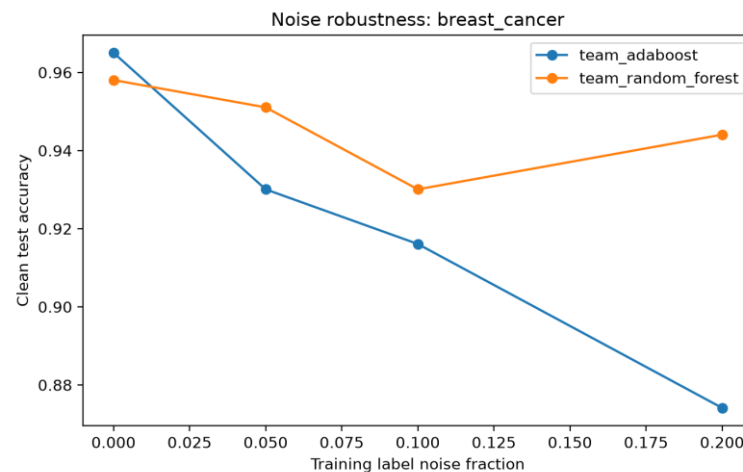
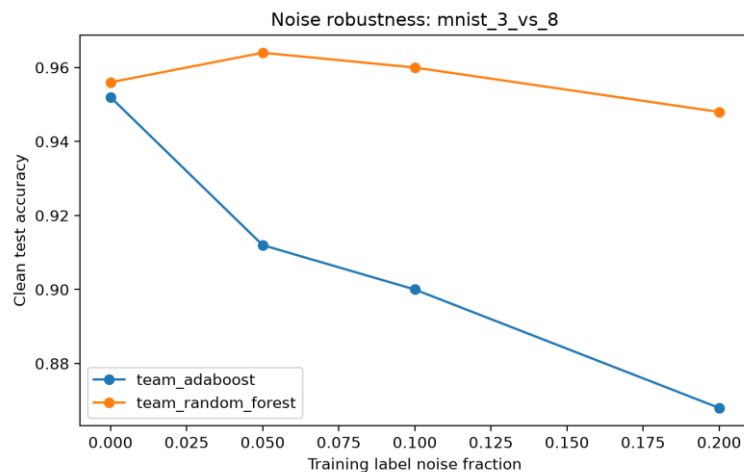
WDBC: both models are strong.

Adult: close performance.

Coverytype: Random Forest is clearly stronger.

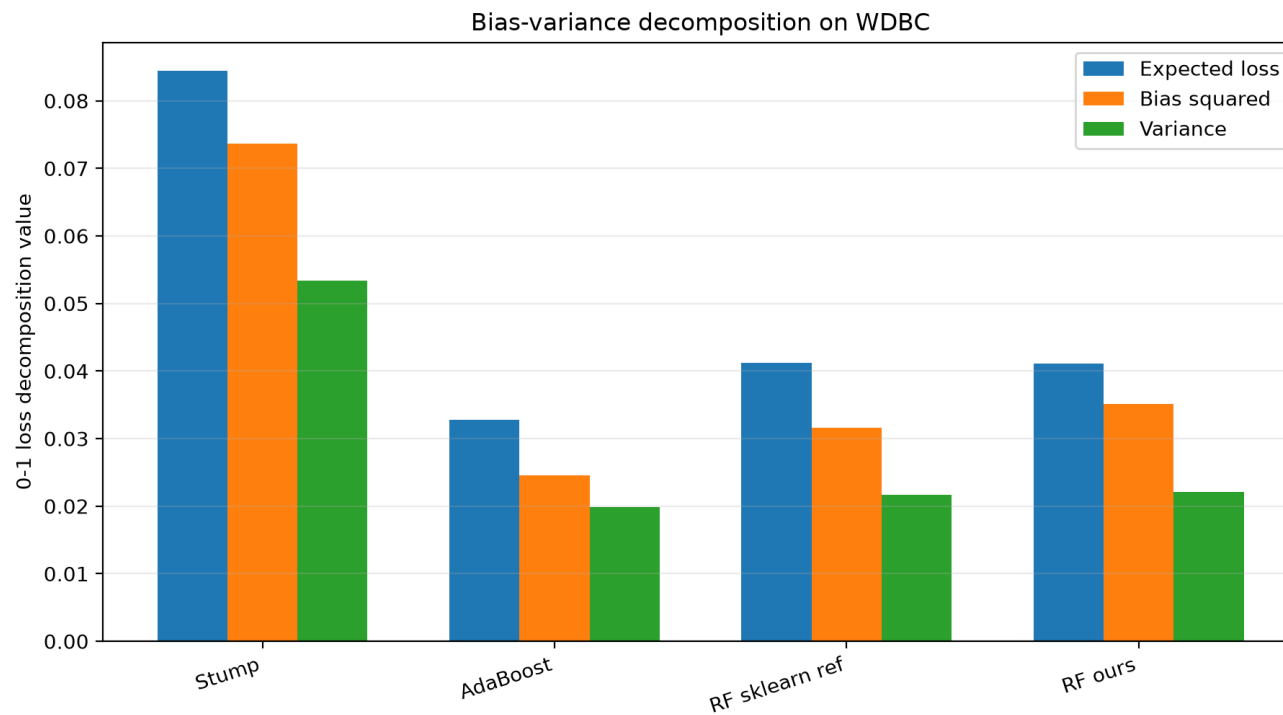
Interpretation: bagging stabilizes heterogeneous tabular structure.

Noise robustness: clean test, corrupted training labels



AdaBoost can up-weight mislabeled samples because it treats them as difficult examples.
Random Forest spreads noise across bootstrap trees and averages it out.
RF is more robust, not immune: extreme noise and severe imbalance still hurt.

Bias-variance decomposition supports the story



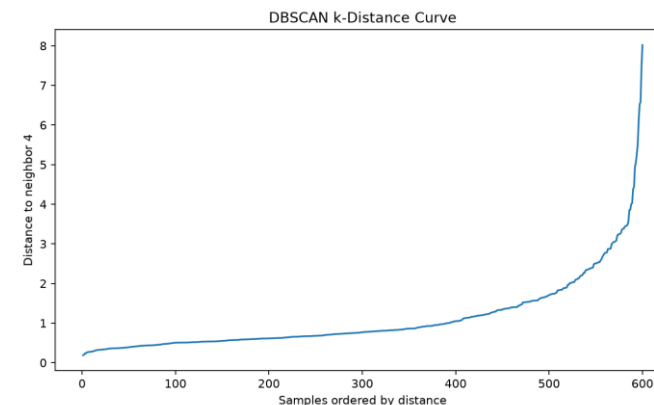
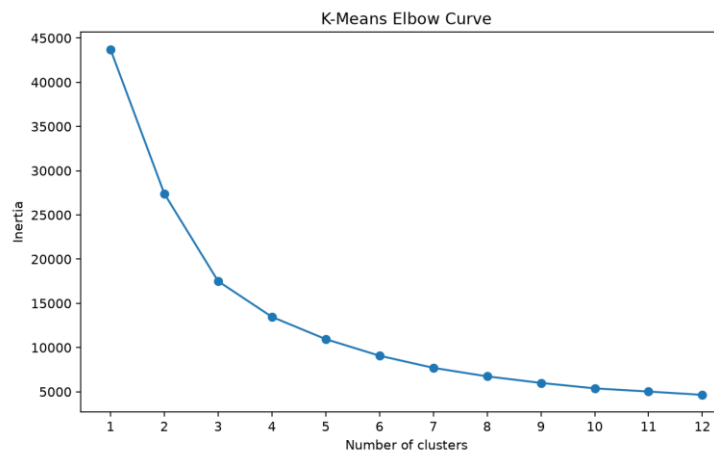
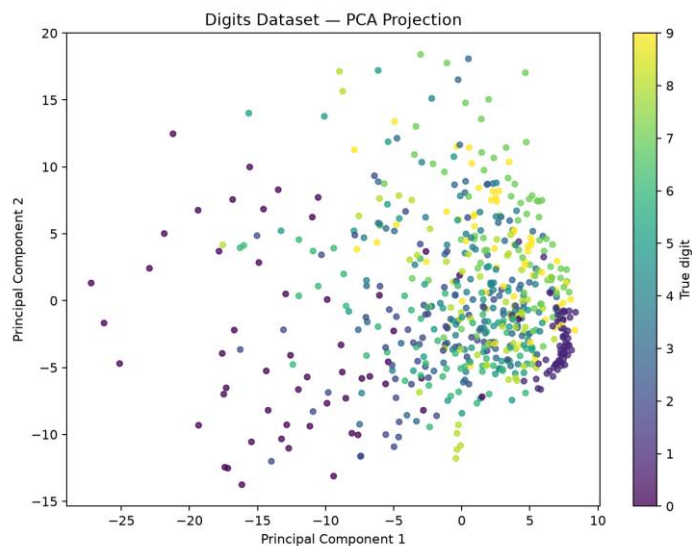
Decision stump: high bias, low flexibility.

AdaBoost: reduces bias by correcting sequential errors.

Random Forest: reduces variance by averaging unstable trees.

Sklearn RF reference follows the same trend as our RF.

Unsupervised diagnostics: structure before classification



PCA shows only partial 2D separation.

K-Means elbow helps pick useful cluster counts.

DBSCAN detects density structure and noise, but is sensitive to eps in high dimensions.

Supervised models still need nonlinear boundaries.

151

tests passed

1

skipped integration

74%

Person 3 coverage

60%

minimum required

RF tests: hard voting, probability sums, rare-class alignment, multiclass labels, OOB validation.

Experiment tests: utility loaders, noise injection, scaling smoke tests, coverage parsing.

Reproducibility: fixed seeds, local data files, root-level results/figures, run_all orchestration.

Data files are prepared by download scripts and should not be committed.

Defense readiness: what each member must defend

İbrahim: Decision Tree, baseline, split logic.

Nihat: Random Forest, OOB, RF experiments, QA.

Fatma: AdaBoost, SAMME/SAMME.R, GBM bonus.

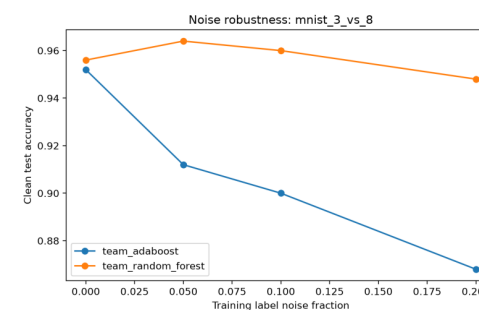
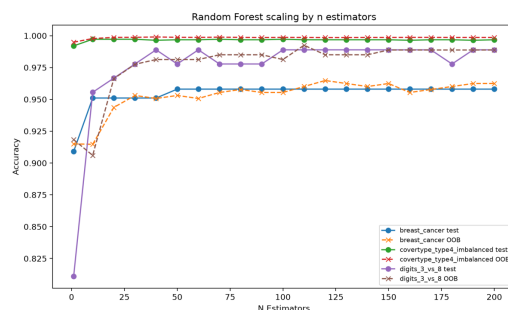
Fidan: PCA, K-Means, DBSCAN, visual diagnostics.

Everyone should derive AdaBoost weight update.

Everyone should explain bootstrap, OOB, Gini/entropy, and bias-variance.

Nihat should be ready for code-level RF questions:

`_fit_single_tree`, `predict_proba` alignment, multiprocessing, and OOB.



Conclusion: choose the ensemble by data condition

Final message

AdaBoost

Best when labels are clean and sequential error correction can lower bias.

Random Forest

Best when robustness, variance reduction, OOB monitoring, and parallel tree fitting matter.

Unsupervised

Helps explain structure, dimensionality, clusterability, and why nonlinear supervised models are needed.

Final answer: Boosting lowers bias; bagging lowers variance. The winner depends on noise, imbalance, dimensionality, and stability.

Thank you